Junyeob Kim

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Education

Seoul National University Mar 2019 – Present
PhD Student in Computer Science and Engineering
Expected graduation: February 2027 | Advisor: Prof. Sang-goo Lee
Seoul, Republic of Korea

Sogang University Feb 2013 – Feb 2019
BS in Computer Science and Engineering
Seoul, Republic of Korea

Current Research

My current research interests lie at the intersection of multimodal understanding, long-context learning, and LLM-based agents. I am particularly interested in how models interpret rich context, retain relevant information, and use it to support coherent reasoning and action over extended tasks. My work spans omni-modal evaluation and next-scene prediction, multilingual long-context mid-training, and memory for web agents. Alongside these research efforts, I am developing an LLM-based Minecraft agent to explore practical collaboration between people and agents.

Multilingual Long-Context Learning *Manuscript under review*

Memory for Web Agents *Manuscript under review*

Professional Experience

NAVER Cloud Oct 2025 – Aug 2026
Foundation Research, AI Residency (WBL Project)

- **Language and reasoning evaluation:** Implemented WBL evaluations; validated Hugging Face log-likelihood compatibility and Qwen2.5-Omni/Emu3 vLLM integration, input formats, and evaluation consistency.
- **Omni-evaluator (separate project):** Contributed to validating the consistency of the evaluation framework and implementing selected components. Related publication: EMNLP 2026 Demo
- **OmniNSP:** Contributed to next-scene prediction research and benchmark development for omni-modal world modeling. Related publication: Findings of EMNLP 2026

Intellisys Sep 2018 – Mar 2020
Research & Software Engineer

- Developed microservices-based backends and Kubernetes-based model serving.
- Researched fashion-item classification, outfit compatibility and recommendation scoring, and item similarity.

Publications

Conference and workshop papers

A Composable Evaluation System for Reproducible Omni-Modal Foundation Model Evaluation
Hodong Lee, Sanghee Park, Dohoon Ryu, Jungwhan Kim, **Junyeob Kim**, Soyoon Kim, Geewook Kim
EMNLP 2026, System Demonstrations [Paper]

Publications Continued

Can Omni-models Imagine the Future? A Diagnostic Benchmark for Multimodal World Modeling
Junyeob Kim, Jihwan Kim, Minha Jhang, Sungwook Jeon, Hwiyeol Jo, Jeonghoon Kim, Sang-goo Lee, Joosung Lee
Findings of EMNLP 2026 [Paper]

Think Just Enough: Leveraging Self-Assessed Confidence for Adaptive Reasoning in Language Models
Junyeob Kim, Sang-goo Lee, Taeuk Kim
Findings of EACL 2026 [Paper]

Aligning Language Models to Explicitly Handle Ambiguity
Hyuhng Joon Kim, Youna Kim, Cheonbok Park, **Junyeob Kim**, Choonghyun Park, Kang Min Yoo, Sang-goo Lee, Taeuk Kim
EMNLP 2024 [Paper]

Adaptive Contrastive Decoding in Retrieval-Augmented Generation for Handling Noisy Contexts
Youna Kim, Hyuhng Joon Kim, Cheonbok Park, Choonghyun Park, Hyunsoo Cho, **Junyeob Kim**, Kang Min Yoo, Sang-goo Lee, Taeuk Kim
Findings of EMNLP 2024 [Paper]

Universal Domain Adaptation for Robust Handling of Distributional Shifts in NLP
Hyuhng Joon Kim, Hyunsoo Cho, Sang-Woo Lee, **Junyeob Kim**, Choonghyun Park, Sang-goo Lee, Kang Min Yoo, Taeuk Kim
Findings of EMNLP 2023 [Paper]

Probing Out-of-Distribution Robustness of Language Models with Parameter-Efficient Transfer Learning
Hyunsoo Cho, Choonghyun Park, **Junyeob Kim**, Hyuhng Joon Kim, Kang Min Yoo, Sang-goo Lee
**SEM 2023* [Paper]

Ground-Truth Labels Matter: A Deeper Look into Input-Label Demonstrations
Kang Min Yoo[*], **Junyeob Kim**[*], Hyuhng Joon Kim, Hyunsoo Cho, Hwiyeol Jo, Sang-Woo Lee, Sang-goo Lee, Taeuk Kim
EMNLP 2022 [*]*Equal contribution* [Paper]

Self-Generated In-Context Learning: Leveraging Auto-regressive Language Models as a Demonstration Generator
Hyuhng Joon Kim, Hyunsoo Cho, **Junyeob Kim**, Taeuk Kim, Kang Min Yoo, Sang-goo Lee
NAACL 2022 Workshop on Large-scale Pre-trained Language Models [Paper]

A Deep Learning based Architecture for Personal A.I. Fashion Stylist Services
Sungchan Park, Jeeseung Han, **Junyeob Kim**, Holim Lim, Sungjun Kim, Jeawoong Jung, Eunyoung Park, Sang-goo Lee, Yuri Lee, Jong-Youn Rha
Artificial Intelligence on Fashion and Textile International Conference (AIFT), 2019

Asynchronous Deep Reinforcement Learning for Vision-Based Autonomous Navigation in Photorealistic 3D Environments
Junyeob Kim
Korea Computer Congress (KCC), 2018. English title translated from Korean.

Publications Continued

Design and Implementation of a Web-Based Training Database System for Lower Limb Rehabilitation Patients
Junyeob Kim, J. Han, J. Yang
Korean Information Science Society Conference (KSC), 2017

Preprint

Investigating the Influence of Prompt-Specific Shortcuts in AI Generated Text Detection
Choonghyun Park, Hyuhng Joon Kim, **Junyeob Kim**, Youna Kim, Taeuk Kim, Hyunsoo Cho, Hwiyeol Jo, Sang-goo Lee, Kang Min Yoo
arXiv:2406.16275, 2024 [Paper]

Teaching Experience

Instructor for listed courses: Prof. Sang-goo Lee.

Database Course, Samsung DS Curriculum 2020, 2021, 2024
Jul – Aug 2020; Jan – Feb and Jul – Aug 2021; Mar – Apr 2024

Recommender System, Seoul National University Mar – Jun 2021

IT Venture Creation, Seoul National University Sep – Dec 2020

Database, Seoul National University Sep – Dec 2019